

Course organization and introduction

ELEC0447 - Analysis of electric power and energy systems

Bertrand Cornélusse

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Overview

1. Course organization
2. Components and structure of an electrical PES
3. Energy outlook for Belgium
4. Insight on renewable and dispersed generation
5. The power balance problem
6. From DC to AC, large interconnections and the come-back of DC

Objectives of this lecture

- ▶ A few words about the organization
- ▶ Show the overall structure of an electric power system
- ▶ Highlight a few important features of power system operation
- ▶ Illustrate those on the Belgian and European systems
- ▶ Present some orders of magnitude it is important to have in mind
- ▶ Introduce some terminology

Adapted from ELEC0014 introduction by Thierry Van Cutsem.

Course organization

The background of the slide features a white central area where the text is located. This area is framed by two large teal-colored triangles that point towards each other from the bottom corners, meeting at a point just below the center of the slide. The overall effect is a clean, modern, and minimalist design.

Course organization

- ▶ Theory lectures (maximum 2 hours)
- ▶ Practice sessions (remainder of the session)
- ▶ Project 1:
 - ▶ Model a simple distribution system in PandaPower and make some simple analyses.
 - ▶ **Individual assessment → bring your laptop**
- ▶ Project 2:
 - ▶ Analyze a system using power flow analyses (power flow solver provided)
 - ▶ **Individual assessment → bring your laptop**
- ▶ Oral exam in January
 - ▶ Theory (list of questions available) and one exercise

ELEC0447 Course Roadmap: From Physics to Grid Analysis

Part I: Component Modeling

- ▶ **L1–2:** Sinusoidal Steady-State & 3-Phase Systems
- ▶ **L3:** Transmission Line π -Model
- ▶ **L4:** Power Transformers & Tap Changers
- ▶ **L5:** Synchronous Generator & Limits

Part II: Power Flow Analysis

- ▶ **L3–4:** Non-linear Power Flow (Newton–Raphson)
- ▶ **L6:** HVDC Links & VSC Converter Control
- ▶ **Projects 1 & 2:** Real-world grid modeling with **PandaPower**

Part III: Operation, Control & Stability

- ▶ **L8:** Frequency Control & Grid Inertia
- ▶ **L9:** Voltage Stability & Reactive Power
- ▶ **L10:** Transient (Rotor Angle) Stability
- ▶ **L11:** Dispatch Center Visit / Operations

Unifying Goal of this Course

Every physical component modeled in Part I directly yields the non-linear algebraic and differential equations you will solve to analyze the real power grid!

The teaching team

- ▶ Bertrand Cornélusse: main contact
- ▶ Francesco Moglia: exercise sessions + project related questions

References

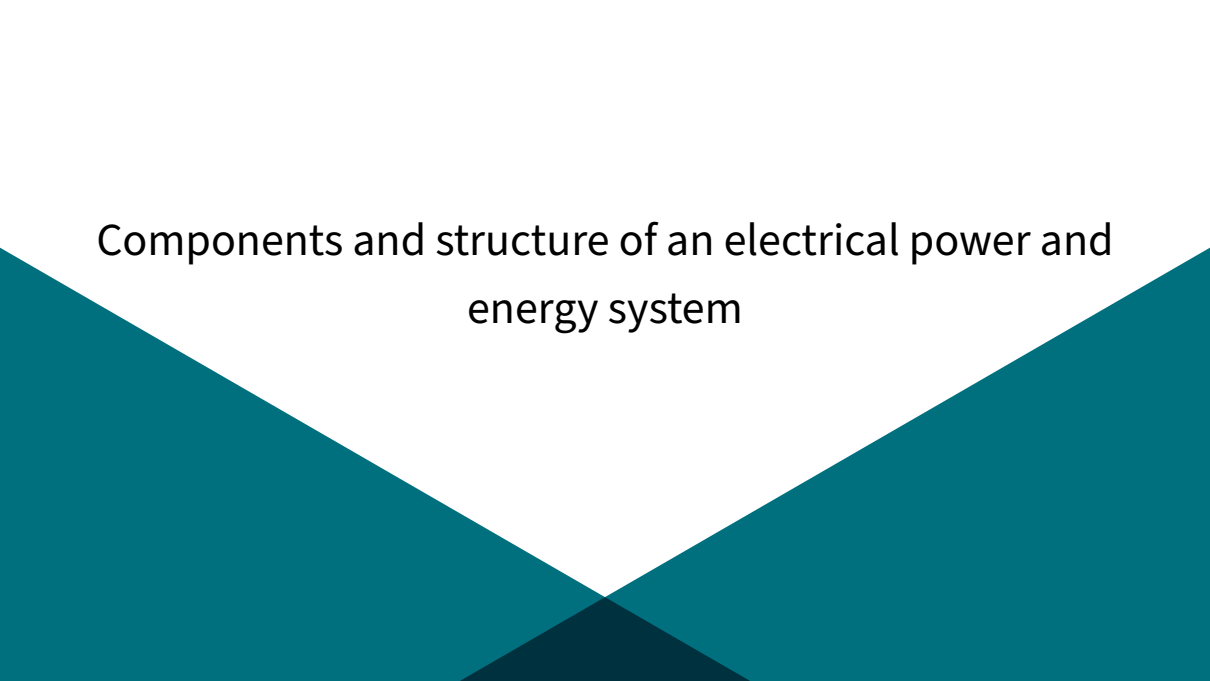
Main reference book:

- ▶ Mohan, Ned. Electric power systems: a first course. John Wiley & Sons, 2012.

Other references:

- ▶ Course notes of ELEC0014 by Pr. Thierry Van Cutsem. (In French)
- ▶ Weedy, Birron Mathew, et al. Electric power systems. John Wiley & Sons, 2012.

Components and structure of an electrical power and energy system

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A large scale system

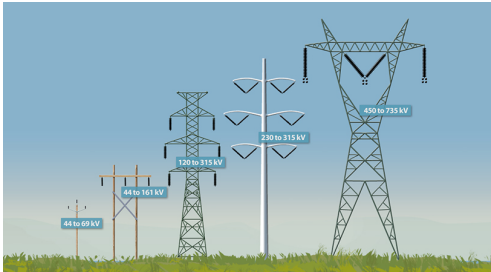
In modern society, electricity has become a “commodity”:

“Commodity: marketable good or service whose instances are treated by the market as equivalent with no regard to who produced them”

But “Behind the power outlet” there is a complex industrial process! Electric energy systems are the largest systems ever built by man:

<https://openinframap.org/>

Transmission and distribution infrastructure: lines, cables, transformers



Source: Hydro-Québec



Source

Generation: centralized power plants to distributed energy resources

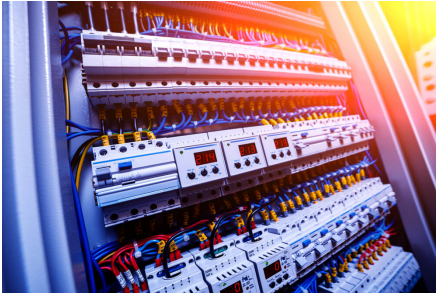


Source: Wikipedia



Source

Protection systems: to eliminate faults



Source

Questions

What is a fault? How can it be detected?
How can it be eliminated? What is the role
of protection relays, circuit breakers, and
disconnectors? Where do we find them?

Switching equipment: substations, circuit breakers, and disconnectors



Source: OSHA

- ▶ **Substations:** nodes connecting circuits at identical or different voltage levels.
- ▶ **Circuit breakers:** capable of interrupting large short-circuit currents under high voltage within a few cycles (20–60 ms).
- ▶ **Disconnectors (isolators):** provide a visible galvanic air gap for safe personnel maintenance.

Real-time measurements

Active and reactive power flows, voltage magnitudes, current magnitudes, energy meters, phasor measurement units



Source

Controllers and operators

Distributed (e.g. in power plant) or
centralized (control center)



Source:

https://eliagrid-int.com/egi_projects/elia-national-control-centre-support/

Markets

- ▶ Forward markets
- ▶ Day-ahead markets
- ▶ Intraday markets
- ▶ Balancing and other ancillary services markets

Question

Do you have an idea why all those markets are needed? What is the difference between them?

There is a dedicated course on these topics (ELEC0018, Energy Markets and Regulation).

Vulnerability to severe environmental disturbances

- ▶ Unlike most human-built infrastructure, thousands of kilometers of grid assets operate exposed to open weather:
 - ▶ Wind storms, ice accretion, conductor galloping
 - ▶ Direct and induced lightning strikes
 - ▶ Wildfires, extreme temperatures, tree contact
- ▶ Mechanical tower collapse and cascading line failures:

Video: Transmission tower mechanical collapse during extreme storm

Low-probability but high-cost failures I

In spite of those disturbances, modern electric power systems are very reliable. Assume a duration of power supply interruption of 0.5 hour / year

$$\text{availability} = (8760 - 0.5) / 8760 = \mathbf{99.9\% !}$$

However, the cost of unserved energy is high

- ▶ average cost used by CREG (Belgian regulator) to estimate the impact of forced load curtailment: 8.3 k€/MWh (source: Bureau fédéral du plan)
- ▶ varies with time of the day : between 6 and 9 k€/MWh

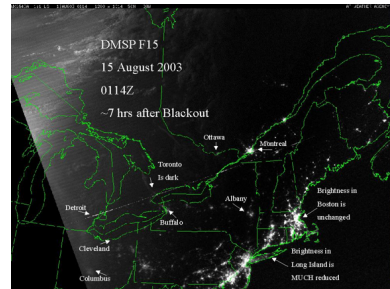
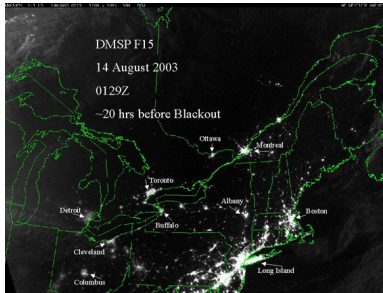
Low-probability but high-cost failures II

- ▶ varies with type of consumer : 2.3 k€/MWh for domestic, much higher for industrial
- ▶ even higher average cost considered elsewhere (e.g. 26 k€/MWh in France !)

Large-scale failures (blackouts) have tremendous societal consequences

- ▶ next two slides: examples of blackouts and their impacts

USA-Canada blackout, August 2003



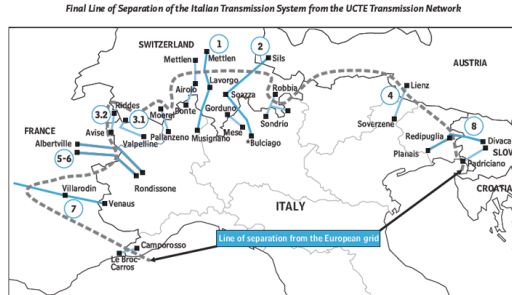
- ▶ 50 million people disconnected initially
- ▶ 61 800 MW of load cut in USA & Canada
- ▶ cost in USA : 4 to 10 billion US \$

- ▶ Canada : 18.9 million working hours lost
- ▶ 265 power plants shut down
- ▶ restoration : from a few hours to 4 days

Source : North American Electric Reliability Council (NERC)

Italian blackout, September 2003 I

- Cascade tripping of interconnection lines → separation of Italy from rest of UCTE system

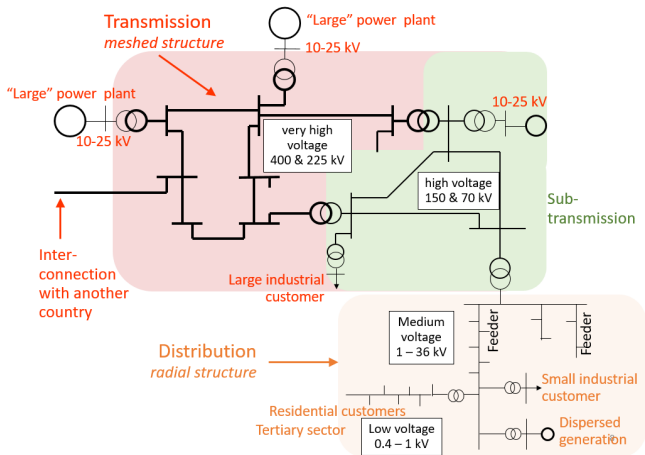


Italian blackout, September 2003 II

- ▶ Deficit of 6.7 GW imported in Italian system → frequency collapsed in Italy
- ▶ 340 power plants shut down, 55 million people disconnected initially - 27 GW lost (blackout occurred during night)
- ▶ Estimated cost of disruption $\approx$ 139 million US \$
- ▶ Restoration time: 15 hours

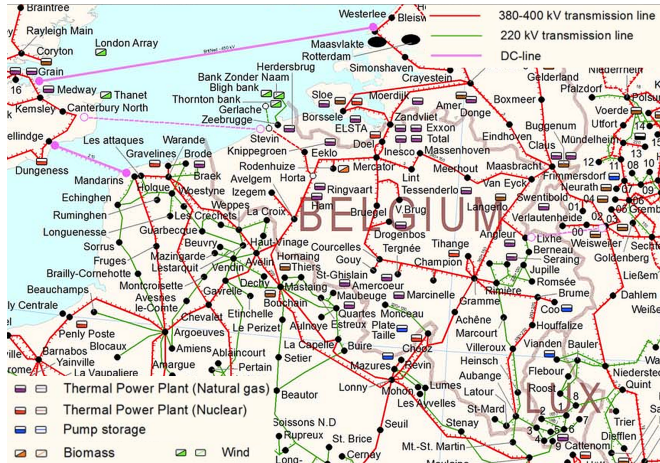
Source : Union for the Co-ordination of Transmission of Electricity (UCTE) which is now part of ENTSO-E

Structure of electric network (case of Belgium)



In Belgium there are 30 and 36 kV underground cable networks, in Brussels and Antwerp areas. These are meshed and play the role of sub-transmission.

400 (and 220 kV) grid in Belgium and interconnections



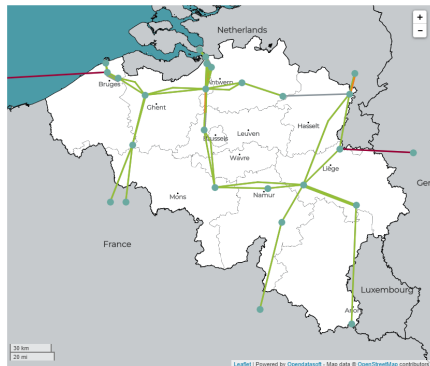
Flows on the Belgian grid

Carte

Reference date: 14/09/2022

Time interval: 

Reference date and time interval: 14/09/2022 09:45 AM



Consult the raw data [Physical flows on the Belgian high-voltage grid](#) | [E.ON Green Data License](#)

Length of network by voltage level and type in Belgium

	Nominal voltage (kV)	Underground cables (km)	Overhead lines (km)	Total (km)
Transmission and sub-transmission				
Very High Voltage	400	49	919	968
	225	47	301	348
High Voltage	150	573	1 981	2 554
	70	301	2 404	2 705
	30 & 36	2 022	60	2 082
	Total	2 990	5 665	8 657
Medium Voltage	Distribution ⁽¹⁾			
	1 ≤ < 30	71 804	5 069	76 873
Low Voltage	< 1 ⁽²⁾	80 480	47 360	127 840
	Total	152 284	52 429	204 713

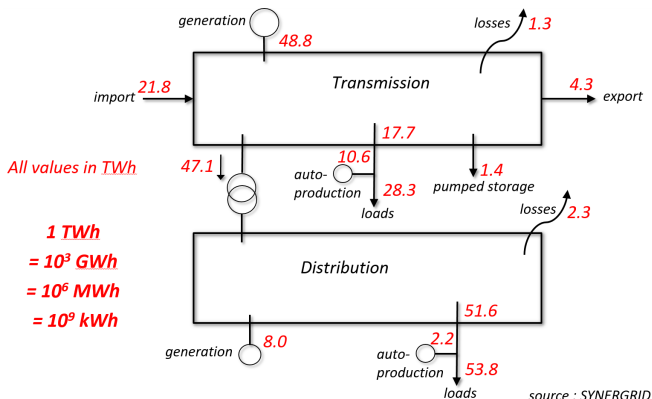
⁽¹⁾ 552 connection points between T & D

⁽²⁾ does not include public lighting

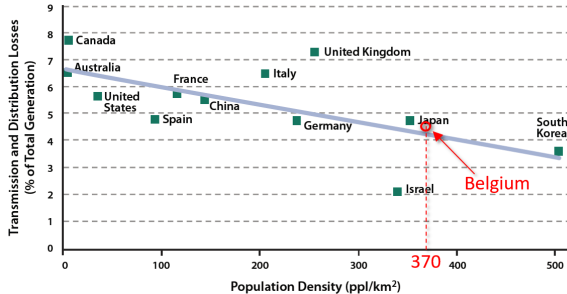
Total number of transformers : 74,990

Source : SYNERGRID, as of December 2008

Electrical energy balance over the year 2018 in Belgium



Network losses

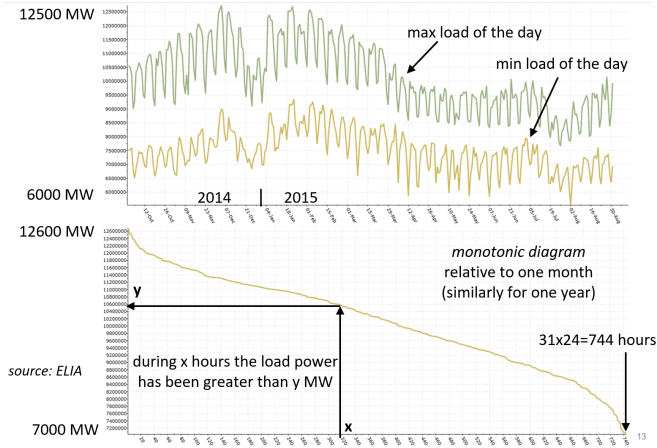


Source: World Bank Development Indicators, <http://data.worldbank.org/indicator>.

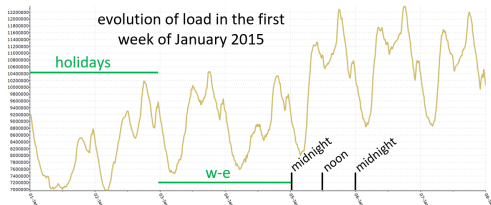
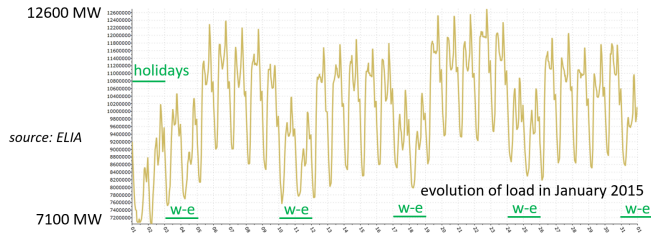
Transporting and distributing electrical energy is an industrial process with a relatively high efficiency!

$$\frac{\text{Transmission \& Distribution losses}}{\text{Production injected in network + import}} = \frac{1.3 + 2.3}{48.8 + 21.8 + 8.0} = 4.6 \%$$

Consumption outlook



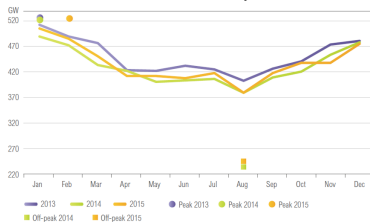
Belgian Electricity Consumption: Seasonal and Weekly Profiles



Peak load on some grids

$$\text{Monthly power in ENTSOe networks} = \frac{\text{energy consumed in the month (GWh)}}{\text{nb days in the month} \times 24}$$

source: ENTSOe



Peak loads
recorded on
the Belgian
transmission
system

Year	Date	Time	Day	Power (MW)
2010	Dec 14	18:00	Tue	14 390
2013	Jan 17	18:00	Tue	13 255
2014	Dec 4	18:00	Thu	12 736
2015	Jan 22	18:00	Thu	12 696
2016	Jan 19	18:00	Thu	12 679
2017	Jan 18	18:00	Thu	13 270
2018	Nov 19	18:00	Mon	13 453

source:
ENTSOe

Energy outlook for Belgium

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Sources of Electrical Energy in Belgium

Category	Energy source	Generation capacity Dec 2018		Energy produced in 2018		Capacity factor
		MW	% total	TWh	% total	%
Nuclear	total	5 919	26.0	27.0	39.0	52
Non renewable non nuclear	gas			22.1		
	others			3.4		
	total	7 680	33.7	25.5	36.9	38
Hydro	pumping stations	1 308		1.0		9
	run-of-river	125		0.3		27
	total	1 433	6.3	1.3	1.9	
Renewable non hydro	wind	3 247		7.1		25
	solar (PV)	3 581		3.5		11
	biomass-biogas	811		3.5		49
	wastes			1.2		
	total	7 664	34.1	15.3	22.1	
TOTAL		22 796	100	69.1	100	

Historical baseline (ENTSO-E)

Live Interactive Check

Open on your smartphone or laptop:

- ▶ **Electricity Maps:** <https://app.electricitymaps.com/zone/BE>
- ▶ **Elia Portal:** <https://griddata.elia.be>

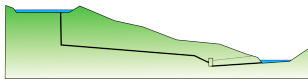
Live observables:

- ▶ What covers demand right now?
- ▶ Is Belgium net exporting or importing?
- ▶ Current carbon intensity (gCO₂/kWh)?

Comments on Generation and Storage

- ▶ **Nuclear:** Historically $\approx 45\text{--}50\%$ of Belgian generation. Significant shifts underway (lifetime extensions vs. decommissioning).
- ▶ **Gas CCGT/CHP:** Crucial for flexible system balancing and peak coverage.
- ▶ **Pumped-Storage Hydro (PSH):**
 - ▶ **Pumping mode:** absorbs excess energy during valley hours.
 - ▶ **Generating mode:** injects at peak demand (75–80% round-trip efficiency).
 - ▶ Delivers rapid primary and secondary frequency reserves (FCR/aFRR) within seconds.

Coo - Trois-Ponts



Coo-Trois-Ponts (1080 MW)

Comments: capacity factor

- Capacity Factor:

$$CF = \frac{\text{energy produced in 1 year (MWh)}}{\text{generation capacity (MW)} \times 365 \times 24(h)}$$

- usually close to 90 % for nuclear, but some Belgian units have high unavailability
- note the low value for solar energy!

Key Energy Trends in Belgium: Generation Mix & Security of Supply

► Nuclear Lifetime Extension:

- Permanent shutdown of Doel 3 (October 2022) and Tihange 2 (February 2023).
- Agreement between the Belgian Government and Engie (2023): **10-year lifetime extension for Doel 4 and Tihange 3** (2 GW operational through 2035).

► Capacity Remuneration Mechanism (CRM):

- Federal capacity market auctions to guarantee security of supply and resource adequacy during the nuclear transition.
- Financing new flexible Combined Cycle Gas Turbines (CCGT, e.g. Flémalle 875 MW) and demand-side response / battery storage.

► Pumped-Storage & Flexibility:

- Coe-Trois-Ponts ongoing modernization (+150 MWh storage capacity) to maximize flexible intraday response.

Key Energy Trends in Belgium: Offshore Wind & Energy Islands

► Offshore Wind Leadership:

- **Zone 1** (Belgian North Sea, 2.26 GW installed) fully commissioned, covering $\approx 10\%$ of total Belgian electricity demand with high capacity factor ($\approx 35\text{--}40\%$).
- **Princess Elisabeth Zone** (Zone 2, up to 3.5 GW additional capacity) currently being developed to triple offshore capacity.

► Princess Elisabeth Island:

- World's first artificial energy island built by Elia in the North Sea.
- Acts as an offshore electrical hub connecting Zone 2 wind farms and landing hybrid interconnectors.

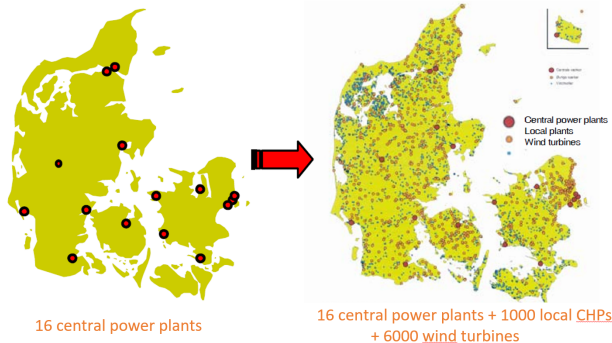
► Cross-Border Interconnections:

- **ALEGrO** (1 GW HVDC to Germany, commissioned 2020) and **Nemo Link** (1 GW HVDC to UK, commissioned 2019) in full commercial operation.
- Future hybrid links: **Nautilus** (UK) and **TritonLink** (Denmark).

Insight on renewable and dispersed generation

From large centralized to small dispersed power plants

Denmark : a country with huge penetration of renewable energy sources

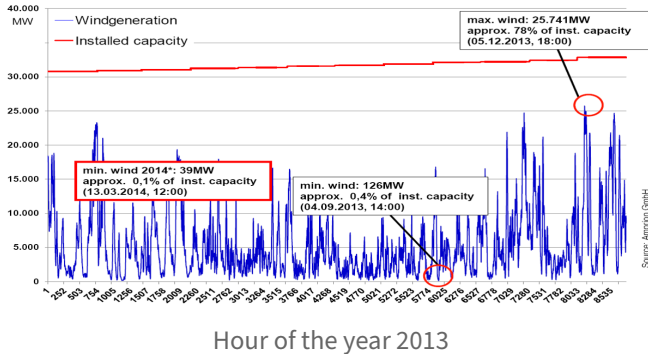


Evolution of Danish power system over the period 1980-2005

(Z. Xu, M. Gordon, M. Lind, J. Østergaard, "Towards a Danish Power System with 50% Wind - Smart Grids Activities in Denmark", IEEEExplore, 2009)

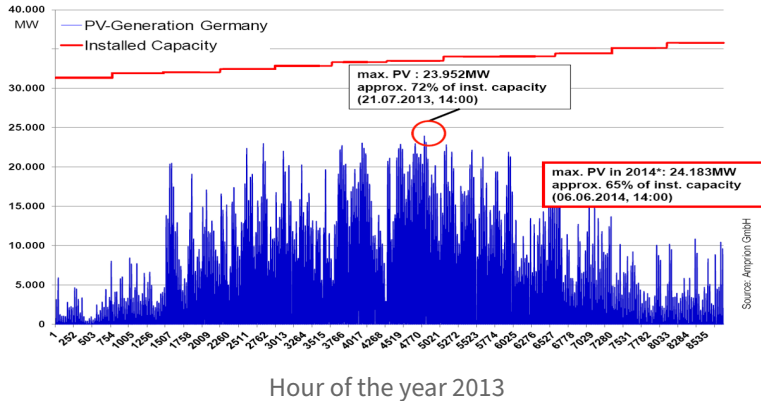
Wind generation potential

<https://globalwindatlas.info/>



PV generation potential

Source



The power balance problem

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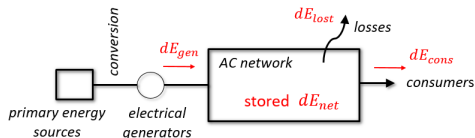
The Power Balance Principle

Conservation of energy over dt :

$$dE_{\text{gen}}(t) = dE_{\text{cons}} + dE_{\text{lost}} + dE_{\text{net}}$$

Dividing by dt , the instantaneous balance is:

$$\mathbf{p}_{\text{gen}}(\mathbf{t}) = \mathbf{p}_{\text{cons}}(\mathbf{t}) + \mathbf{p}_{\text{lost}}(\mathbf{t}) + \mathbf{p}_{\text{net}}(\mathbf{t})$$



- ▶ $p_{\text{cons}}(t)$: Total consumer demand (stochastic, fluctuating).
- ▶ $p_{\text{lost}}(t) = \sum R_k i_k^2$: Joule losses in lines and transformers.
- ▶ $p_{\text{net}}(t)$: Rate of change of electromagnetic energy in L and C .

Energy Storage and the AC Network

► Can we store electrical energy in the AC grid itself?

- In sinusoidal steady state, power in an inductor or capacitor alternates every quarter cycle (2ω).
- In balanced three-phase operation:

$$\sum_{k=a,b,c} p_{L,k}(t) = 0, \quad \sum_{k=a,b,c} p_{C,k}(t) = 0 \implies \mathbf{p}_{\text{net}}(\mathbf{t}) \equiv \mathbf{0}$$

- **Fundamental consequence:** *No net energy can be stored within the AC transmission network!*
- **Storage requires conversion to other energy forms:**
 - **Gravitational:** Pumped-storage hydro (e.g. Coe: 1080 MW, > 75% efficiency).
 - **Electrochemical:** Battery Energy Storage Systems (BESS) (lithium-ion, NaS, flow batteries) expanding exponentially for fast frequency response (< 1 s).
 - **Chemical:** Electrolysis → hydrogen / e-fuels (seasonal storage).

The Need for an Energy Buffer: Power System Inertia

► The Balancing Challenge:

- Consumer demand fluctuates continuously.
- Primary mechanical energy conversion (steam/gas valves, hydro gates) takes several seconds to react!

► The Rotating Mass Buffer (Inertia):

- Massive rotors of synchronous generators provide an instantaneous kinetic energy buffer:

$$E_{\text{kin}} = \frac{1}{2} J \omega^2$$

- Rotor kinetic energy is typically equivalent to 2 to 5 seconds of nominal generator power!
- Generation deficit ($p_{\text{gen}} < p_{\text{cons}} + p_{\text{lost}}$) → rotors decelerate → **grid frequency drops.**
- Generation surplus ($p_{\text{gen}} > p_{\text{cons}} + p_{\text{lost}}$) → rotors accelerate → **grid frequency rises.**

► Modern Frontier (Inverter-Based Resources):

- Wind, solar, and batteries have no intrinsic physical inertia.

From DC to AC, large interconnections and the
come-back of DC

Network: from early DC ...

End of 19th century : Gramme, Edison devised the first generators, that produced Direct Current (DC) under relatively low voltages Impossibility to transmit large powers with direct current:

- ▶ $\text{power} = \text{voltage} \times \text{current}$
 - ▶ if the voltage cannot be increased, the current must be
 - ▶ but $\text{power lost} = \text{resistance} \times \text{current}^2 \rightarrow$ big waste of energy
 - ▶ and large sections of conductors required $\rightarrow$ expensive and heavy
- ▶ Hard to interrupt a large DC current (no zero crossing), for instance after a short-circuit

Network: ... to present high-voltage AC

Changing for Alternating Current (AC)

- ▶ voltage increased and lowered thanks to the transformer
- ▶ standardized values of frequency : 50 and 60 Hz (other values used at a few places)

Larger nominal voltages have been used progressively

- ▶ up to 400 kV in Western Europe
- ▶ up to 765 kV in North America
- ▶ experimental lines at 1100 kV or 1200 kV (Kazakhstan, Japan, etc.)

Motivations for large interconnections

- ▶ Mutual support between partners to face the loss of generation units
- ▶ Each partner would have to set up a larger “reserve” if it were to operate isolated
- ▶ Larger diversity of energy sources available within the interconnection
- ▶ Allows exploiting complementarity of nuclear, hydro and wind power plants
- ▶ Allows partners to sell/buy energy, to create a large electricity market.

Operational Challenges of Large AC Interconnections

- ▶ **Fault Propagation:**

- ▶ If one partner is unable to isolate a major contingency, disturbances and cascade trips propagate across the synchronous zone.

- ▶ **Loop Flows and Parallel Paths ("Wheeling"):**

- ▶ Power flows according to Kirchhoff's laws (impedances), not commercial contracts!
- ▶ Transactions between country A and B induce unscheduled flows through neighboring countries C and D (see next slide).

- ▶ **Inter-Area Electromechanical Oscillations:**

- ▶ Poorly damped low-frequency oscillations (0.1–0.5 Hz) where rotor angles in one region swing against another region.

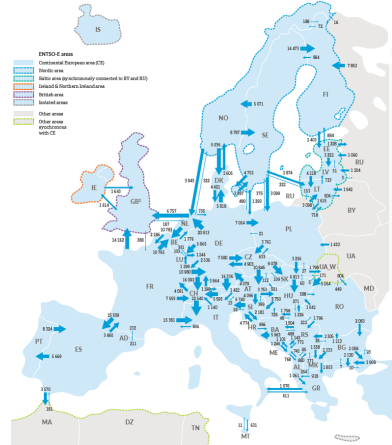
- ▶ **Frequency & Power Quality Harmonization:**

- ▶ Requires strict, shared grid codes and primary frequency reserves across all member TSOs.

European networks

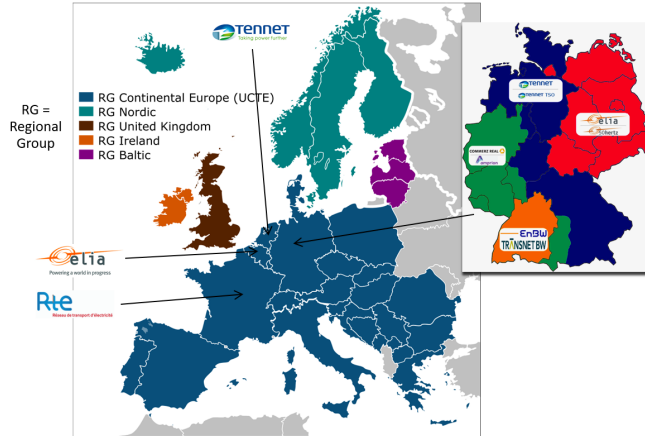
ENTSO-E : European Network of
Transmission System Operators for
electricity

41 Transmission System Operators (TSOs)
from 34 countries



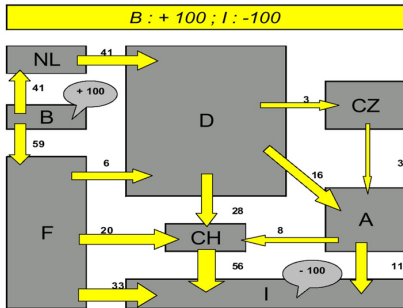
Energy flows in 2018 (in GWh)

The synchronous grids of Europe



Example of paths followed by power due to a transaction

Paths taken by a production increment of 100 MW in Belgium covered by a load increase of 100 MW in Italy (variation of losses neglected):



The Comeback of Direct Current: Modern HVDC

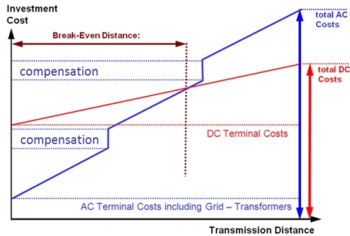
► Power Electronics Revolution:

- Transition from thyristors (Line Commutated Converters, LCC) → IGBT-based **Voltage Source Converters (VSC)** and **Modular Multilevel Converters (MMC)**.
- Independent, four-quadrant control of active power P and reactive power Q .
- Capable of connecting to weak AC grids and providing **black-start** restoration.

► Key HVDC Transmission Domains:

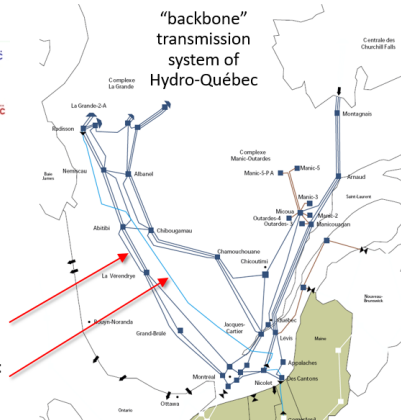
- Bulk power transmission over long overhead distances ($> 600\text{--}800\text{ km}$).
- Subsea and underground high-voltage cables ($> 50\text{ km}$).
- Controlled asynchronous interconnectors between independent AC synchronous zones.

Transmission over longer distances through overhead lines



dark blue: AC transmission at 735 kV

light blue: HVDC link :
1018 km $\pm$ 450 kV 2000 MW



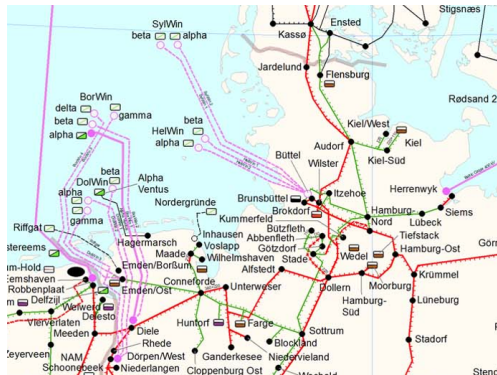
Transmission through Submarine and Underground Cables

► Why HVDC for cables > 50 km?

- High capacitance of insulated cables produces continuous AC reactive charging current:
$$I_c = \omega CU.$$
- Beyond ≈ 50 km, charging current exhausts the thermal capacity of the cable!
- In DC: $dU/dt = 0 \implies$ no steady-state capacitive charging current.

► Operational Interconnectors in Belgium:

- **Nemo Link:** 1000 MW, ± 400 kV subsea link to UK.
- **ALEGrO:** 1000 MW, ± 320 kV underground link to Germany.



North Sea offshore wind HVDC connections
(TenneT / ENTSO-E)

Asynchronous Grid Coupling via HVDC Back-to-Back Links

- ▶ **Networks with different nominal frequencies:**
 - ▶ Coupling of 50 Hz and 60 Hz systems in Japan
 - ▶ Coupling of Brazil (60 Hz) with Argentina (50 Hz)
- ▶ **Networks with identical frequency but asynchronous:**
 - ▶ Decoupled to avoid propagating severe dynamic instability
 - ▶ Continental Europe (ENTSO-E) with Great Britain, Nordic, and Baltic systems
 - ▶ Eastern, Western, and ERCOT interconnections in North America

